

Reviewer Pack — Hypergeometric Function Coefficients in Communication Comple...

Entry 42e0f2d2085c · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Hypergeometric Function Coefficients in Communication Complexity via q -Difference Operators

Entry ID: 42e0f2d2085c **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-31 15:16:23 UTC

1. Conjecture

Field A (mathematical branch): q -Series Analysis **Field B** (complexity object): Communication Complexity

Statement:

For any given k -ary communication protocol P , the number of distinct non-zero hypergeometric function coefficients in its associated q -difference operator representation is upper-bounded by $\Omega(k \log n)$, where n is the input size.

Rationale (proposer's reasoning):

q -Series Analysis has been used to study various combinatorial structures. The proposed mapping from communication protocols to q -difference operators might reveal a deeper connection between algebraic properties of these operators and the complexity of communication protocols, potentially leading to novel insights in communication complexity theory.

Taxonomy category: q -Series Analysis $\times$ Communication Complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 117f89d56f91c966

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The number of distinct non-zero hypergeometric function coefficients in the q -difference operator representation of a k -ary communication protocol is upper-bounded by $\Omega(k \log n)$, where n is the input size.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_q_difference_operator(k, n):
        # Placeholder for q-difference operator generation logic
        return [random.randint(0, 1) for _ in range(n)]

    def count_non_zero_hypergeometric_coefficients(operator):
        return sum(1 for x in operator if x != 0)

    k = random.randint(2, 5) # Randomly choose k between 2 and 5
    n = random.randint(5, 40) # Randomly choose n between 5 and 40

    operator = generate_q_difference_operator(k, n)
    count = count_non_zero_hypergeometric_coefficients(operator)

    metric_value = count
    conjecture_holds = count >= k * math.log(n)
    counterexample = "" if conjecture_holds else f"k={k}, n={n}, count={count}"

    return {
        "metric_name": "Non-zero hypergeometric coefficients",
        "metric_value": metric_value,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] if len(sys.argv) > 1 else [random.randint(2, 97) for _ in range(5)]

    results = []
    for seed in seeds:
```

```

trial_result = run_trial(seed)
print(f"TRIAL: {trial_result}")
results.append(trial_result)

if all(result["conjecture_holds"] for result in results):
    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value) ** 2 for result in results) / len(results))
    support_fraction = len([result for result in results if result["conjecture_holds"]]) / len(results)
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
elif any(not result["conjecture_holds"] for result in results):
    first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")
else:
    print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

conjecture_holds': False, 'counterexample': 'k=3, n=14, count=6'}
TRIAL: {'metric_name': 'Non-zero hypergeometric coefficients', 'metric_value': 7, 'instances_tested': 10}
TRIAL: {'metric_name': 'Non-zero hypergeometric coefficients', 'metric_value': 11, 'instances_tested': 10}
TRIAL: {'metric_name': 'Non-zero hypergeometric coefficients', 'metric_value': 9, 'instances_tested': 10}
TRIAL: {'metric_name': 'Non-zero hypergeometric coefficients', 'metric_value': 5, 'instances_tested': 10}
TRIAL: {'metric_name': 'Non-zero hypergeometric coefficients', 'metric_value': 14, 'instances_tested': 10}
TRIAL: {'metric_name': 'Non-zero hypergeometric coefficients', 'metric_value': 19, 'instances_tested': 10}
TRIAL: {'metric_name': 'Non-zero hypergeometric coefficients', 'metric_value': 14, 'instances_tested': 10}
TRIAL: {'metric_name': 'Non-zero hypergeometric coefficients', 'metric_value': 17, 'instances_tested': 10}
TRIAL: {'metric_name': 'Non-zero hypergeometric coefficients', 'metric_value': 21, 'instances_tested': 10}

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8879df4d.py", line 63, in <module>
    print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")
    ^^^^^
NameError: name 'result' is not defined. Did you mean: 'results'?

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results show that for some protocols, the maximum count of distinct non-zero hypergeometric function coefficients exceeds the upper bound of | next: Investigate the specific protocols and conditions under which this counterexample occurs to better understand the limitations of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	17243
2	preregistration	ollama_remote	glm4:latest	0	0	9210
3	novelty	ollama_remote	glm4:latest	0	0	11236
4	novelty	ollama_remote	glm4:latest	0	0	8420
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10977
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	22724
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8590
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6785
9	judge	ollama_remote	glm4:latest	0	0	19899

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 115084 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/42e0f2d2085c.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/42e0f2d2085c.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/42e0f2d2085c.tar.gz` (if generated)